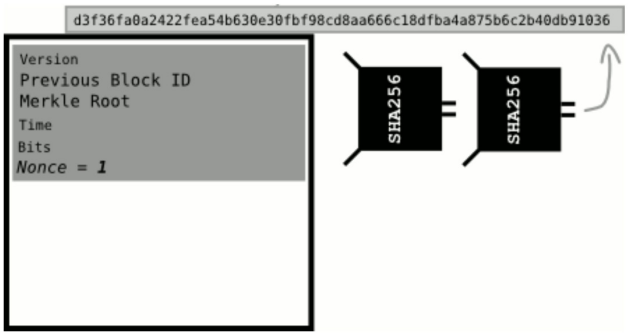
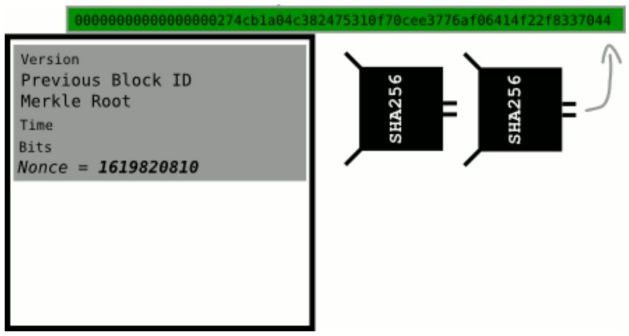
8.4 Rehashing the Hashes *- No Pun Intended*

How long does it take miners to discover a valid hash? And how quickly can they modify the nonce values during their calculation process?

In the example below, it took a miner 1619820810 iterations to find a hash value with the required number of zeros. The miner who succeeded in finding the right nonce value added the block to the blockchain, forming a secure and unalterable part of the chain.





The **hash rate** is a measure of the computational power of the network and the **speed at which miners can make nonce calculations**, which are used to find the correct block hash.

The more computational power a miner has, the faster they can make these calculations, which can give them an advantage in the mining process. However, as the network's hash rate increases, the difficulty of mining new ***bitcoin*** also increases, making it more challenging for all miners to find the correct **block hash**.

- Just like athletes upgrade their equipment to run faster, Bitcoin miners invest in specialized computer hardware to increase their **hash rate** and mine blocks more efficiently. The more resources they invest in, the better their chances of reaching the finish line first.
- Hash rate can be compared to the speed of the runner. The more powerful the miner's machine, the higher the hash rate, and the faster they can mine. However, just like in a race, having a high speed does not guarantee a win if the difficulty target has been adjusted to a higher level. Miners must constantly upgrade their equipment and improve their hash rate to stay ahead of the competition and have a chance of mining a block and winning the race.

The process of finding the correct **hash value** by changing the **nonce**, is what is called **mining**!